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Alnylam Pharmaceuticals: Building Value from the IP Estate

A number of things make our situation really interesting. The first of which is, the science all came together at around the same time in 2000–2001 when Tom Tuschl discovered the siRNA structure and its activity. Then he put the RNA into human cells and showed he could silence genes. That was a singular event. At that point, though people recognized that the technology was a long way from actually being clinically viable, the whole notion that one could target any gene by design was incredibly appealing.

So Tom and the team did two things: they patented it, but more interestingly, even knowing that it was way too early to start a company, they founded Alnylam. They wanted to be the first movers and they wanted to be in the lead for developing this technology. So the company was started, and now we have what we believe is a ground-changing technology. But we know that it's going to take five, seven, 10, 12 years to get the technology into the market . . . to get approval as a therapeutic. And that's going to require an enormous investment.

— John Maraganore, CEO, Alnylam Pharmaceuticals

John Maraganore, CEO of Alnylam Pharmaceuticals, was reflecting on the IP “estate” that **Alnylam** had built. His firm and its partner, the Max Planck Institute for Biophysical Chemistry in Göttingen, Germany, had **filed a lawsuit against the Whitehead Institute, the Massachusetts Institute of Technology (MIT) and the University of Massachusetts (UMass)** over an aspect of patents that was not often litigated: the **prosecution¹ of a patent that the institutions had jointly filed and directed by Whitehead**. The patent, originally filed on March 30, 2000 as *provisional* patent application USSN 60/193,594 and later converted into a *non-provisional* application on March 30, 2001 under patent application USSN/09/821,832, became known as the “Tuschl I” patent. **It was widely regarded as an important patent in the emerging field of RNA interference (RNAi) because of its early date and precedence over much subsequent work.**

The Tuschl I patent was filed based on work done on the drosophila fruit fly, and it was jointly owned by the Max Planck Institute, MIT, the Whitehead Institute, and UMass. By prior agreement Whitehead led the prosecution of the patent. **Even though it had not yet been issued, the partners had begun licensing the patent in anticipation of its issuance. Alnylam had a license,** as did Sirna Therapeutics (now wholly owned by Merck) and CytRx Corporation. (CytRx subsequently spun out its RNAi assets into a company called RXi Pharmaceuticals.) While Whitehead, MIT, Max Planck, and

¹ The language of patenting has many specific terms, which we have italicized on their first use in this case. These terms are defined in **Exhibit 1**.

Professor Willy Shih and HBS Doctoral Candidate Sen Chai prepared this case. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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Alnylam would have preferred that Tuschl I be licensed exclusively to Alnylam, UMass licensed its interest in the Tuschl I applications to Sirna and RXi.

Subsequently, Tom Tuschl and two other inventors at Max Planck made an RNAi breakthrough discovery in human cells. The invention was assigned to Max Planck, which filed a patent application in Europe on December 1, 2000 (EP 00126325.0) known as “Tuschl II,” which was assigned exclusively to Alnylam. In addition to Tuschl I and Tuschl II, there were six other patents that the Alnylam founders and many others viewed as “must haves” in the area of intellectual property (IP) for RNAi therapeutics (see **Exhibit 2** for all eight frontier patents). Alnylam had secured licenses or some form of agreement covering all eight of these patents to ensure its freedom to operate.

But what was the interplay in the prosecution of Tuschl I and Tuschl II? Why were the erstwhile partners squabbling? Maraganore, in the circumspect style befitting a pending litigation, observed:

We are very confident in our overall patent estate. However, in a case like this, you get disagreements that can occur, particularly when you have intellectual property elements where there are multiple owners of the original invention, like multiple universities, because the inventors happened to be in different academic institutions. So you have a fragmentation of sort of the ownership interests.

DNA, RNA, Sense, and Antisense

Deoxyribonucleic acid (DNA) is a molecule that contains the genetic instructions essential to the functioning of living organisms. Like a set of biological blueprints, it carries a code that is used to construct proteins and other components of cells. Chemically, DNA consists of two long strands of simple building blocks called nucleotides attached to a backbone that together are wound into the shape of a double helix. Attached along each backbone are four different molecules called bases (abbreviated “A,” “T,” “C,” and “G”), which uniquely complement each another. Thus, “A” and “T” always pair, as do “C” and “G.” It is the sequence of these bases, arranged as triplets or “codons,” that makes up the genetic code. Thus, a codon might be A-T-G, whose complementary base pairs would be T-A-C (see the top section of **Exhibit 3** for a diagram).

The process that dictates how genetic information embedded in DNA is transcribed into ribonucleic acid (RNA) and then translated into proteins is the central dogma of molecular biology. It was first articulated by James Watson and Francis Crick in 1953. Genetic sequence information is transformed into proteins and other biological components via three types of information transfers: transcription—the process of copying a sequence of DNA into RNA; translation—the process of transforming RNA sequences into proteins; and DNA replication—the step in which genetic information is transmitted from parent to progeny (refer again to **Exhibit 3**).

Though most people who have had even the briefest exposure to biology understand that DNA and genes are the basis for the genetic code and heredity, it was many years before it became clear that RNA’s role was central to the whole process. The insight that single strands of complementary nucleotides could bind to strands of mRNA (“antisense” strands) and mess up the whole translation mechanism led to some revolutionary experiments.

The Discovery of RNA Interference and its Application in Therapeutics

The first observation that antisense strands of RNA technology had the potential to selectively control gene expression dated back to 1977.ⁱ The idea was that if one knew the gene sequence that

coded for a particular protein, one could use a complementary antisense strand to selectively target and “down regulate” a specific gene. This meant one could target and block genes that were thought to be central to the development of specific diseases.

Some of the earliest experiments that informed this thinking were conducted by Richard Jorgensen and Carolyn Napoli, who were trying to create more intensely colored petunias. Using techniques for the genetic engineering of plants, they tried to introduce specific segments of DNA that would cause the overproduction of color pigments in the flowers. But instead of getting a deeper color, the plants lost most of their pigmentation. They got pure white or patterned flowers instead!ⁱⁱⁱ The expression of the gene was *suppressed* rather than enhanced. The really surprising thing was that the suppression was reversible, and some of the white flowers reverted to wild purple flowers. This phenomenon became known as “co-suppression” – the idea that a piece of DNA that was related to a specific gene could be involved in suppressing the expression of that gene.

Following this discovery, there was a flurry of activity in using single-stranded complementary, synthetic DNA strands as a way to silence specific genes, a technology that became known as “antisense.” Stan Crooke, CEO of Isis Pharmaceuticals, filed a series of patents covering different chemical modifications that scientists could make to stabilize the DNA segments so that they could reach their targets inside cells. These patents became known as the “Crooke patents.” Isis also filed patents that claimed chemical “building blocks” to make DNA and RNA segments with drug-like properties.

Antisense advanced slowly in the field of nucleic acid therapeutics until 1998, when Andrew Fire, a developmental biologist then at the Carnegie Institution in Washington (and doctoral student under Phillip Sharp of the MIT Whitehead Institute) and Craig Mello from the UMass Medical School made a major breakthrough in the understanding of the underlying mechanism of this gene silencing process.ⁱⁱⁱ They introduced various forms of long double-stranded RNA (dsRNA) molecules into the nematode worm *Caenorhabditis elegans* and observed that it was dsRNAs that were actually responsible for inducing the RNAi gene-silencing phenomenon. Unlike synthetic single-stranded antisense strands, dsRNAs exist naturally in the cells of many species, so they aren’t likely to be rejected by the body or pose toxic side effects. Rather, dsRNAs would harness the cell’s own machinery to silence its genes. What was revolutionary was that Fire and Mello provided a reproducible way to silence specific genes in simple organisms that were commonly used to study genetics. They were awarded the 2006 Nobel Prize in Physiology or Medicine for this discovery. They also filed a provisional patent, “Genetic Inhibition by Double-Stranded RNA,” in 1997, and were granted U.S. Patent 6,506,559, which issued on January 14, 2003.^{iv}

The Fire and Mello technique led to numerous experiments in other organisms, but the widespread belief was that it would not work in mammalian cells, where the infusion of long double-stranded RNA would trigger an interferon response that killed those cells. Research at two European labs changed this notion dramatically. Roland Kreutzer and Stefan Limmer at the University of Bayreuth experimented with short fragments of dsRNA, hoping to trigger the same RNAi response achieved with long dsRNA. They were able to show that these short fragments worked in mammalian cells. Kreutzer and Limmer filed a series of key patents around their discovery and left Bayreuth to start a company called Ribopharma AG in 2000, the first company formed to focus on the development and commercialization of RNAi therapeutics.

At about the same time, Tom Tuschl, who had recently completed a postdoctoral fellowship with Phillip Sharp, returned to the Max Planck Institute and independently conducted experiments by infusing small RNA molecules (that had short sequences of base pairs) directly into mammalian cell cultures, including human cell cultures. He discovered that small double-stranded RNAs or small

interfering RNAs (siRNAs) of 19–23 base-pair length were in fact the mediators of RNAi gene silencing. These discoveries led to a historic *Nature* paper^v and also to the filings of the Tuschl II patents. **Exhibit 5** pictorially describes the natural RNA interference process.

RNA interference can be initiated in cells using two methods, either by the “expressed” (endogenous) or “delivered” (exogenous) routes. The discoveries described above corresponded to the delivered route because the RNA molecules were supplied exogenously. Benitec, a small Australian firm whose research was led by the principal instigator, Michael Graham, was the leader in the expressed system route and held key patents in this area. Its process involved inserting a DNA construct into a cell, triggering the production of double-stranded RNA that was then immediately cleaved into small interfering RNA (siRNA) segments, which then entered the cellular RNAi pathway. A timeline for key RNAi discoveries, patent filings, and patent grants is shown in **Exhibit 4**.

RNA interference (induced by dsRNA) harnesses a natural pathway that acts upstream from where traditional medicines work. Instead of blocking the physiological function of proteins or mimicking its effect (agonist drugs) or targeting enzymes by regulating the rate of chemical reaction (inhibitor or activator drugs), RNAi-based drugs turn off genes at the RNA level even before translation into proteins occurs.

The excitement over RNAi was around the potential for an entirely new therapeutic paradigm. Many, if not most, diseases involved overexpression of a gene. Since RNA interference could target a specific gene and silence it, this meant that genetically based diseases might be treated simply by silencing the offending gene. This was a revolutionary concept -- it was the “holy grail” of genomics—the idea that one could directly translate a genetic sequence into a drug. This potential RNAi therapeutic mechanism is depicted in **Exhibit 6**.

Practically speaking there was still the challenge of delivering the appropriate siRNA to where it was needed in the body. This problem had to be solved before widespread commercialization of RNAi-based drugs could occur. The furthest along RNAi-based drugs targeted eye, lung, and liver diseases, thanks to the ease of their delivery. Eye and lung medications were directly applied externally, while the natural body pathway ultimately took such molecules to the liver following injection. If the challenge could be addressed, for example, by delivering siRNA to a diseased cell through nanoparticles, RNA interference could become a game-changing technology in pharmaceuticals.

Recognizing the Therapeutic Potential of RNAi

We developed the strategy at the beginning to develop the leading product-focused company and importantly, to consolidate the intellectual property landscape. The reason we did the IP consolidation strategically was to avoid the fragmentation of value that occurred in the settings of other big platform technologies and to be the leading company for pharma collaborations . . . [B]etween June of 2002 and roughly the end of 2003, in a dogged, persistent, calculated fashion, we literally had a road map of all the key inventions in the field, who the inventors were, what were the academic institutions where the licenses were being held, and we negotiated all these different license grants – license agreements – with these universities. In one case we acquired a company in July of 2003, Ribopharma, because they had some critical patents. So it was a very explicit strategy to bring it all together. It was at a time in that technology where frankly there was not as much data to support its potential . . . [T]here was excitement about it, but that was not the hard-core data saying that this was going to become a new drug class. So we had the luxury of a relatively non-competitive situation, as well as the persistence and the conviction that we wanted to consolidate IP this way, which would give us the whole lion’s share of the value.

— John Maraganore

Alnylam was founded when the therapeutic potential of RNAi was still far from certain. Though the mechanistic understanding of gene silencing was immensely promising, there was still uncertainty and a lot of R&D yet to be done on how siRNA could be delivered to cells effectively. The company founders formulated a strategy that had four major legs:

- Bifurcate the fields of use for RNAi into (1) laboratory reagents and (2) therapeutic uses.
- Assemble a patent portfolio (the “patent estate”) that anybody who anticipated therapeutic use RNAi would have to license.
- License the portfolio selectively to raise as much cash as possible.
- Invest the proceeds from licensing in R&D to develop the necessary delivery mechanisms for RNAi to have therapeutic use and advance RNAi therapeutics to the market.

Once the decision to bifurcate fields of use was made, Alnylam quickly licensed a number of companies to sell siRNA laboratory reagents, but it held back on the therapeutic side. The thinking at the time was to encourage scientists to use siRNA technology in their own research and win validation of the approach in those scientists’ minds. They based their strategy on what had happened to antisense technology. Crooke and Isis had built a solid IP estate around antisense, but because it was not widely available, many in the scientific community did not experiment with it, and consequently never built any faith in the approach.

Assembling the Patent Estate

Alnylam’s founders had a substantial cumulative experience in biotechnology, and they were collectively quite concerned about how the patent landscape might develop (see **Exhibit 7** for a list of the founders). In particular, the IP landscape of monoclonal antibodies caused them great concern, and they did not want RNAi to develop that way. Maraganore explained:

Going back to monoclonals ... we knew that in the case of monoclonals there was the Winter-Boss patent owned by the Medical Research Council in the U.K., there was the Cabilly patent owned by Genentech, there was the Queen patent owned by PDL (PDL BioPharma), a company called Protein Design Labs. . . . [T]hose are three examples. And each of them had a different sliver. Then there were a whole bunch of patents owned by Abgenix, and a whole bunch of patents owned by Medarex, so anybody who was developing a monoclonal antibody at the time would have to license this one, and then this one, and then that one.

The analysis of the patent landscape led to the identification of the eight *frontier patents* in RNAi as well as others that might have subsidiary importance. The company embarked on a systematic approach of securing either ownership or freedom to practice under each one that it did not already control. In the case of the Kreutzer-Limmer patents, it bought the company (Ribopharma AG) that owned the patent family. (A summary of its deals is shown in **Exhibit 8**.)

Maraganore (and many others) felt that the estate was pretty strong. While the Fire and Mello patent was truly at the frontier of research, it was broadly available to anyone. The Tuschl II patents were frontier in the application to mammalian cells, and Alnylam had exclusive licensing control over those. Alnylam had a cross-license to the Crooke patents in the *field of use*, and also had a cross-license with Benitec. The Kreutzer-Limmer patent family faced repeated challenges in the European

Patent Office, but at least Alnylam controlled those as well, regardless of the outcome. The only “leak,” if it could be called that, was Tuschl I, which had not yet been granted by the United States Patent and Trademark Office (USPTO). The Max Planck Institute had agreed in 2001 to allow cross-claiming of *priority dates* between Tuschl I and II and the use of experimental data from Tuschl II in the Tuschl I application, but only if it did not harm the Tuschl II *patent family*. MIT and Whitehead were supposed to let the USPTO know that the *claims* of Tuschl I did not depend on this supporting data. But by 2004, Max Planck had already started asking Whitehead to remove all Tuschl II material from Tuschl I.

Regardless of issues surrounding Tuschl I, Alnylam was able to demonstrate the value of its patent estate through the ultimate test: the willingness of other companies to license the portfolio, and for substantial sums of money.

Licensing the Portfolio to Raise Non-dilutive Funding, and Funding R&D

What’s been wonderful so far with our strategy is that we have been able to raise all this non-dilutive funding capital from pharma partnerships through the licensing of our intellectual property in a way that is giving us the ability to invest in our pipeline and has allowed us to be able to frankly escape going to the capital markets to raise the capital. We always had that option available to us, but frankly the cost of that capital is pretty high. I’ll give you the numbers; since the beginning – we’re an eight-year-old company – we’ve raised in total \$950 million; actually this year we will probably hit \$1 billion. We only raised \$250 million of the \$950 million in the sale of equity to venture capitalists or to the public markets, and the remaining \$700 million has been raised all through non-dilutive cash payments from partnerships, which is all foundationally based on licensing technology and patents to the third-party pharmaceutical companies.

– John Maraganore

In order to build a top-tier, science-based, biotechnology therapeutic company, it was important not only to aggressively demonstrate scientific advancement through patents and peer-reviewed journal articles, but also to ensure a pipeline of multiple programs, either proprietary or partnered. As Maraganore observed, “Obviously with what we are doing, we still have to prove that the science will deliver the sort of products the way the monoclonals have delivered . . . We are hopeful!”

With the strong intellectual property estate Alnylam had built throughout its eight-year history, it was able to selectively license its IP, providing non-dilutive research funding. Of the \$950 million Alnylam had raised to date, more than \$700 million came from non-dilutive licensing fees.

Furthermore, the company had to be able to fund the ever-increasing cost of therapeutic research. Maraganore reflected:

If you look historically at Genentech, Amgen, Chiron, Biogen, and Genzyme, if you look at their cumulative investment from the point at the beginning all the way to having cash flow positive EPS, it costs them anywhere from \$1 to \$2 billion of investment capital over 10 to 20 years to get there . . . Going through the math and going back to the \$900 million, and maybe this year \$1 billion that we have raised to date, we’ve invested a half-billion dollars into our technology cumulatively over time. We have about \$400 million in the bank, but the interesting thing is if you look at our long-range plan, from where we are today to the point where we are cash flow positive based on product earnings, we might have to invest another \$1 billion.

One of Alnylam's earliest licensing deals was signed in September 2005, when the company formed an alliance with Novartis aimed at discovering, developing, and commercializing RNAi therapeutics for a defined number of gene targets exclusively selected by Novartis. In addition, Novartis acquired an option to a non-exclusive license to Alnylam's platform. In July 2007, Alnylam struck one of its biggest licensing deals yet with the Swiss pharmaceutical giant Hoffman-La Roche (Roche),² providing the Swiss drug maker a non-exclusive license to Alnylam's technology platform for developing RNAi therapeutics. Then, in May 2008, Alnylam licensed the Japanese drug maker Takeda Pharmaceutical, granting Takeda broad, worldwide, non-exclusive access to and enablement with Alnylam's RNAi therapeutics platform technology and intellectual property in the fields of oncology and metabolic disease. Takeda also obtained the right to expand the number of therapeutic fields in the future. The alliance included the transfer of platform technology from Alnylam to Takeda, and a collaboration and cross-license of delivery technologies. Through Regulix Therapeutics, a joint venture spinoff between Alnylam and Isis, Alnylam also forged license agreements in the microRNA arena with U.K.-based GlaxoSmithKline (in April 2008) and in June 2010 with France-based Sanofi-Aventis.

Not only did these licensing deals establish a value from Alnylam's IP estate (which indirectly placed a valuation on the company itself), they also provided Alnylam the opportunity to get grant-back licenses to discoveries from licensees as well as access to other IP platforms from the alliances. Licensing thus acted as a hedge in case Alnylam's own research program failed, as it enabled Alnylam to cherry-pick other partners' successful programs and potentially obtain partial ownership and favorable economics. Maraganore had one cautionary note:

This can't be done in every single company [as licensing IP potentially enables competitors], but if you have a very broad platform technology where in theory there are hundreds and hundreds of potential drugs that can come out of it, you can provide people the enablement from a freedom-to-operate standpoint and get the value of that from a licensing standpoint without worrying about it, by giving a license grant to a partner.

One other aspect of this licensing strategy was that Alnylam had to constantly monitor the research landscape in search of controlling IP from other potentially game-changing inventions in the burgeoning subfields of RNAi. Alnylam had to have a significant portion of its research capacity focused on unresolved scientific issues, such as the delivery of RNAi-based drugs. Maraganore elaborated, "We have about a \$100 million gross operating budget for this year. We will spend about \$40 million on delivery technology, and we will spend about \$60 million on our pipeline, clinical pipeline [which consists of] the cost of clinical trials that are needed to advance drugs to the marketplace."

The Lawsuit

Although Alnylam was unwilling to talk about the lawsuit while it was still pending, much information could be gleaned from publicly available documents, including a blog^{vi} dedicated to following the case.

The complaint that Alnylam and Max Planck filed against MIT, the Whitehead Institute, and UMass asserted that the dispute arose from "the prosecution of two sets of patent applications that

² Hoffman-La Roche owned Genentech.

are directed to inventions that came out of separate lines of research, performed by separate groups of investigators, in the field of RNA interference”: Tuschl I and Tuschl II. The complaint went on to explain that the Tuschl I application had been jointly assigned by the four inventors to Max Planck, Whitehead, MIT, and UMass, and that Whitehead was responsible for the prosecution of Tuschl I. Max Planck filed its Tuschl II European patent application after the Tuschl I non-provisional application was filed. In March 2001, in preparing to convert the Tuschl I non-provisional application into a provisional application, Whitehead, through prosecution counsel, asked whether Max Planck would agree to allow the Tuschl I and Tuschl II applications to be combined and prosecuted as one application. Max Planck refused to allow the applications to be combined but did allow Whitehead to incorporate significant descriptions of the Tuschl II invention into the specification of the Tuschl I application. The complaint asserted that the defendants then began using the description of the Tuschl II invention to support claims in the Tuschl I application—particularly relating to the use of RNAi to treat humans—that detracted from the commercial value of Tuschl II. (The first four pages of the complaint are shown in **Exhibits 9** and **10**.)

Attorneys for UMass and Whitehead stated in court that the inventive matter of Tuschl II was not invented by the Tuschl II inventors, and that this matter was obvious and inherent in Tuschl I.

The problems likely originated in 2003 when UMass independently licensed its rights to Tuschl I to two direct competitors of Alnylam: Sirna Therapeutics and CytRx. UMass had no right to license Tuschl II, as it was assigned exclusively to Max Planck and licensed to Alnylam. But UMass, Sirna Therapeutics, and CytRx had an economic incentive to try to convince the USPTO to issue a Tuschl I patent that would encompass the Tuschl II invention, thereby preventing a Tuschl II patent from issuing. Sirna Therapeutics, which Merck & Co. later acquired in December 2006 for \$1.1 billion, subsequently argued to Alnylam that it had the rights to use the description of the Tuschl II invention in the specification of the Tuschl I application to support claims that would include the Tuschl II invention.

Some would argue that the root cause of this dispute could be traced to the fact that Alnylam did not acquire UMass’s rights to Tuschl I at any cost. Douglas M. Fambrough, a venture investor in Sirna Therapeutics, suggested that this was the case: “Alnylam let UMass get away, and Sirna licensed Tuschl I from UMass. [It is] an unacceptable failure in securing your IP when starting a company.” However, the intricacies of Tuschl I’s assignment decision were quite complex, as all inventors on the patent traced their roots back to the lab of Phillip Sharp (a cofounder of Alnylam) at MIT. Tom Tuschl had been a postdoctoral fellow working for Sharp at Whitehead, who subsequently moved to Germany for a job at the Max Planck Institute. Phil Zamore, who was a doctoral student in Sharp’s lab, left for an academic position at UMass. David Bartel was another researcher and close collaborator with Sharp at MIT.

Thus, Max Planck, MIT, the Whitehead Institute, and UMass all ended up as co-owners on the first Tuschl patent application (Tuschl I), but Max Planck retained sole rights to the subsequent work that Tuschl did in Germany, including the critical discovery of the optimal structure of siRNAs and their activity in mammalian cells (Tuschl II).

“How Badly Do I Need a License?”

Do you try to exclude everybody? The intellectual property is pretty good . . . There are some people who argue, “I can go around the edges . . . I can put a little nucleotide here.” Tuschl has these, we bought Ribopharma to get one of the earlier patents, it’s all about reduction to practice. Big Pharma, if they want to fight you, they have the money to do it, so they can just grind you away with litigation!

— John Maraganore

Merck paid dearly to acquire Sirna Therapeutics. Did Merck think it had acquired sufficient freedom to practice in RNAi drug development under Tuschl I and Sirna's IP estate? No doubt Merck's interests were aligned with Whitehead and UMass in the current litigation. Was Merck also the major force behind the appeal of several of the Kreutzer-Limmer patents by the European Patent Office? Did Merck really need Tuschl II to develop RNAi-based drugs? Was it just trying to secure its IP position?

Merck could still make significant strides in the delivery stage for RNAi therapeutics. If it were to develop key IP in this area, it could potentially gain important negotiation leverage over the clinical application of the technology and it would transform RNAi therapeutics from an early-stage research-based biotechnology into a product-based multibillion-dollar therapeutic market. Then, if it still felt it needed a license to Tuschl II, it could be in a very different negotiating position.

Was there one solution to the delivery problem? Maraganore reflected on the possibilities:

Historically you would anticipate "no." Once it is proven that this is a really a whole new therapeutic paradigm, people will invest billions of dollars on delivery technology. So it is possible there will be multiple approaches. I would anticipate that there will be more than one intellectual "key." You see how aggressive Alnylam is on siRNAs; we're just as aggressive on delivery! At least we have a way to pay for it.

I think there is a whole landscape of products, clearly more than a start-up like us can ever develop by ourselves . . . [W]ith all those development time lags and that patent clock running, we're going to have to be there and be fast, and be clever continuously for the next 20 years to take advantage of this. If someone else develops a billion-dollar product out of this, that's great! There's enough space in cancer if this technology works . . . [Y]ou could spend the rest of a pharmaceutical company's life delivering drugs for cancer.

There was another potential move: Alnylam could license Merck (owner of Sirna). Was that a good idea or a bad idea? Or did it only depend on the price? Then again, the price would almost certainly be higher if the current trial went to completion and Alnylam and Max Planck prevailed.

Exhibit 1 Key Patent Terminology

Term	Meaning
Citations	A list of references that are believed to be prior art, and which therefore help to “narrow” the application. May be made by the applicant or the examiner.
Claims	Section of a patent that technically defines the scope of protection granted by the patent or sought in a patent application. The main claim defines the invention in its broadest form, and dependent claims describe additional features.
Continuation	Second or subsequent applications for the same invention that were claimed in a prior application that is in part or directly derived from the original invention. Only applicable in the U.S.
Disclosure	The public revealing of the details of an invention—described in the patent application. If the invention is disclosed prior to the filing of an application, it is unpatentable. Some inventors/companies deliberately disclose inventions so that no one else can patent them.
Field of Use Limitation	A provision in a patent license that limits the scope of what a licensee can use the patent for by defining a specific area where the license is applicable
First to File	The inventor who is first to file in this system is entitled to the patent. This is the standard for countries adhering to the Trade-Related Aspects of Intellectual Property (TRIP) guidelines, which includes most countries except the U.S.
First to Invent	The inventor who is first to invent in this system is entitled to the patent. This necessitates keeping good records, with a witnessed laboratory notebook.
Frontier patent	A patent that takes precedence by virtue of being early in a field and describes an invention that others will build upon.
Infringement	Occurs when another party unlawfully makes, uses, offers for sale, or sells, without permission, one or more claims of the invention protected in a patent during the term of the patent and within the country in which the patent is issued.
License	The means by which a patent owner gives another person or entity the right to do things that, absent a license, would constitute infringement. Note that patents are exclusionary rights, so a license waives the exclusion, but does not guarantee the right to use the invention.
Patent family	A set of patents filed in multiple countries to protect a single invention.
Patentable invention	A new way of solving a problem that is useful and not obvious to one skilled in the field.
Prior art	Previously used or published technology that is relevant to the invention.
Priority date	The date of first filing of the patent application.
Prosecution	The process that describes the interaction between patent applicants and their counsel with the patent office regarding a patent or the application for a patent. Patent prosecution can be divided into pre-grant and post-grant. During pre-grant prosecution, the patent applicant negotiates with the patent office for the grant of a patent, while during post-grant prosecution, the two sides deal with amendments and opposition issues related to the patent.
Provisional Application	A U.S. patent application filed under 35 U.S.C. §111(b). It allows filing without a formal patent claim, oath or declaration, or any information disclosure (prior art) statement. It provides the means to establish an early effective filing date in a later filed non-provisional patent application filing. It offers a lower cost first filing vehicle.
Publication	Outside the U.S., patent applications are published 18 months after the priority date. In the U.S., prior to November 29, 2000, applications were kept confidential until grant. After that date, they were published 18 months after the priority date (there are exceptions).
Validity/Invalidity	Whether a patent should have been at all allowed by the PTO, whether asserted patent claims are sufficiently new or non-obvious in light of the prior art, or whether other requirements of patentability have been met.

Source: Compiled by casewriters.

Exhibit 2 RNA Interference: Eight Frontier Patents

Patent	Inventors	Patent No.	Status	Date	Assignee	Licensees	Coverage
Fire and Mello	Andrew Fire, Stephen Kostas, Mary Montgomery, Lisa Timmons, SiQun Xu, Hiroaki Tabara, Samuel E. Driver, Craig C. Mello	6,506,559	Issued	January 14, 2003	Carnegie Institution of Washington	Unrestricted, broadly licensed	A process is described for introducing an RNA into a living cell to inhibit gene expression of a target gene in that cell. The process may be practiced <i>ex vivo</i> or <i>in vivo</i> . The RNA has a region with a double-stranded structure. Inhibition is sequence-specific and uses double-stranded RNA 25 or more nucleotides in length.
Tuschl I	Thomas Tuschl, Phillip D. Zamore, Phillip A. Sharp, and David P. Bartel	U.S. 2002/0086356 A1, application pending	In prosecution	March 30, 2001	Whitehead Institute for Biomedical Research	Alynham Pharmaceuticals, Sima Therapeutics, CytRx Corp.	Inhibition of target genes using double-stranded RNA between 21 and 23 nucleotides in length. Included overhangs, chemical modifications, and data from mammalian cells.
Tuschl II	Thomas Tuschl, Sayda Mahgoub Elbashir, Winfried Lendeckel	7,056,704	Issued	June 6, 2006	Max-Planck-Gesellschaft zur Foerderung der Wissenschaften e.V.	Alynham Pharmaceuticals	Double-stranded RNA (dsRNA) induces sequence-specific post-transcriptional gene silencing in many organisms by a process known as RNA interference (RNAi). Using a <i>Drosophila in vitro</i> system, demonstrated that 19–23 nucleotide short RNA fragments are the sequence-specific mediators of RNAi.
Tuschl II	Thomas Tuschl, Sayda Mahgoub Elbashir, Winfried Lendeckel	7,078,196	Issued	July 18, 2006	Max-Planck-Gesellschaft zur Foerderung der Wissenschaften, e.V.	Alynham Pharmaceuticals	Double-stranded RNA (dsRNA) induces sequence-specific post-transcriptional gene silencing in many organisms by a process known as RNA interference (RNAi).
Crooke	Stanley T. Crooke	6,107,094	Issued	June 6, 1997	Isis Pharmaceuticals, Inc.	Alynham Pharmaceuticals	Described chemical modifications to stabilize RNA.
Crooke	Stanley T. Crooke	5,898,031	Issued	April 27, 1999	Isis Pharmaceuticals, Inc.	Alynham Pharmaceuticals	Described chemical modifications to stabilize RNA.
Benitec	Michael Wayne Graham	6,573,099	Issued, then pulled for reexamination	June, 3, 2003	Benitec Australia, Ltd.		Genetic constructs for delaying or repressing a target gene using expressed system.
Kreutzer-Limmer	Roland Kreutzer, Stefan Limmer	EP 1,144,623	Issued in 2002 to Ribopharma, but revoked		Ribopharma (purchased by Alynham)		Broad claims related to RNAi mediating double-stranded RNA (dsRNA) molecules of 15–21 base pairs (bp) in length. In opposition proceedings, claims reduced to blunt-ended molecules joined by a chemical linker. In the appeal instance, the TBA held that the disclosure of the application as filed does not allow a claim to dsRNA molecules of 15 to 21 bp because support can only be acknowledged for a range of 15 to 49 bp and an exemplary 21 bp long dsRNA. This was not considered an adequate basis for selecting a sub-range of 15 to 21 base pairs. Hence, the subject matter was rejected under Article 123(2) EPC. However, a related case, the “945 patent” was recently upheld in Appeal Proceedings before the EPO and covers dsRNA with a length of 15–49 base pairs.

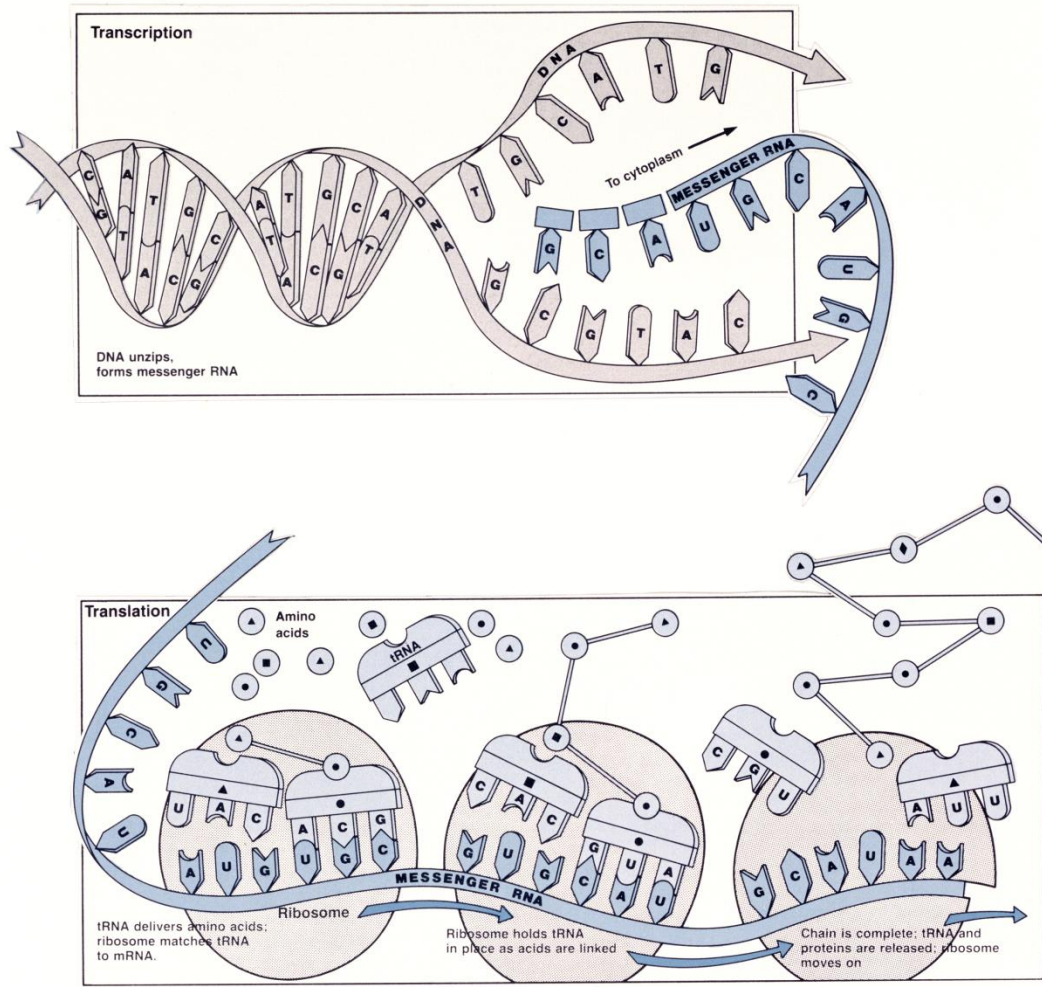
Source: Compiled by casewriters from multiple published sources.

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Taught by Prof. Dr. Bart Leten, from 12-Feb-2024 to 30-Jun-2024. Order ref F501327.

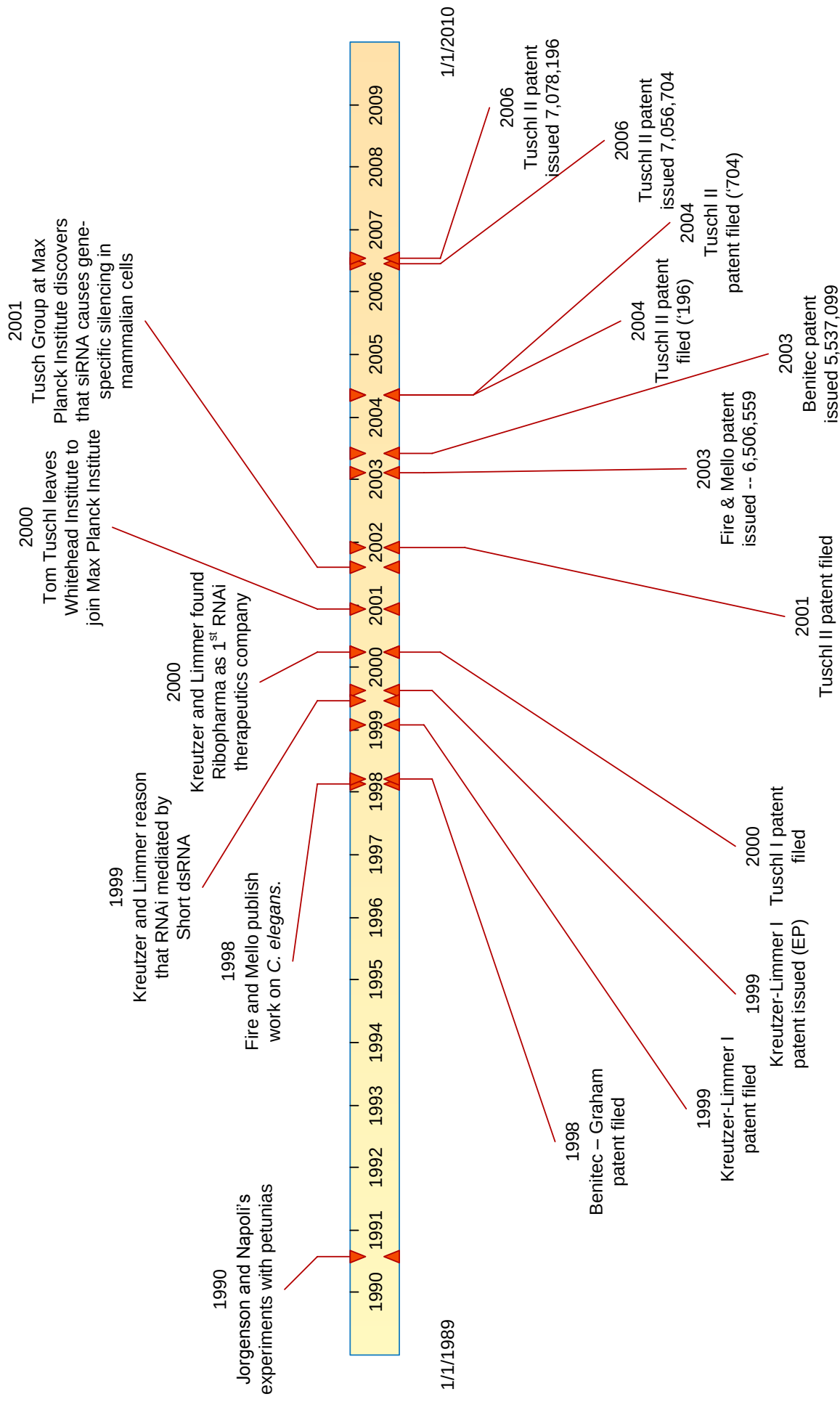
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Exhibit 3 DNA Transcription Process



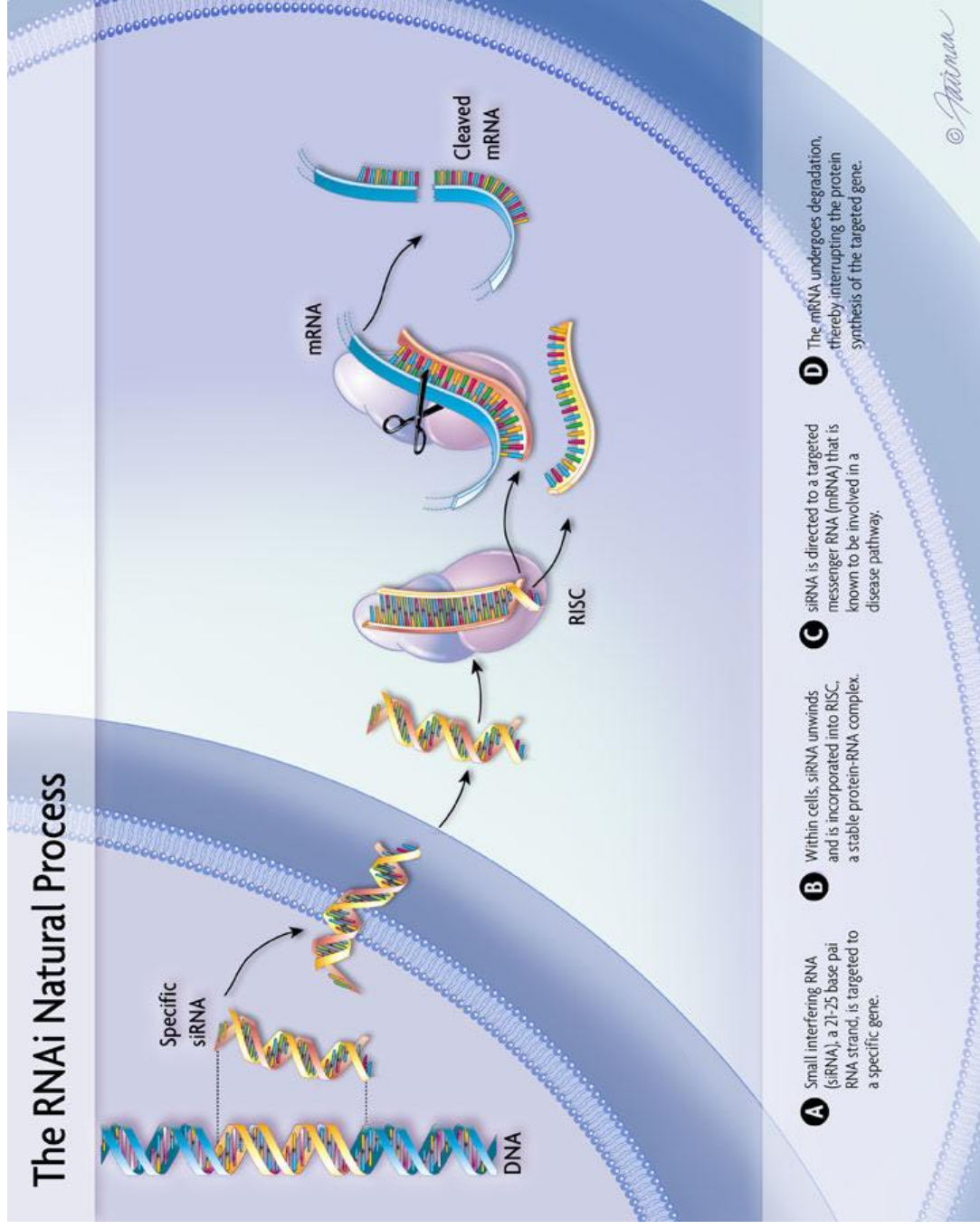
Source: Lawrence Berkeley National Lab, used with permission.

Exhibit 4 RNAi Key Discovery and Patent Time Line



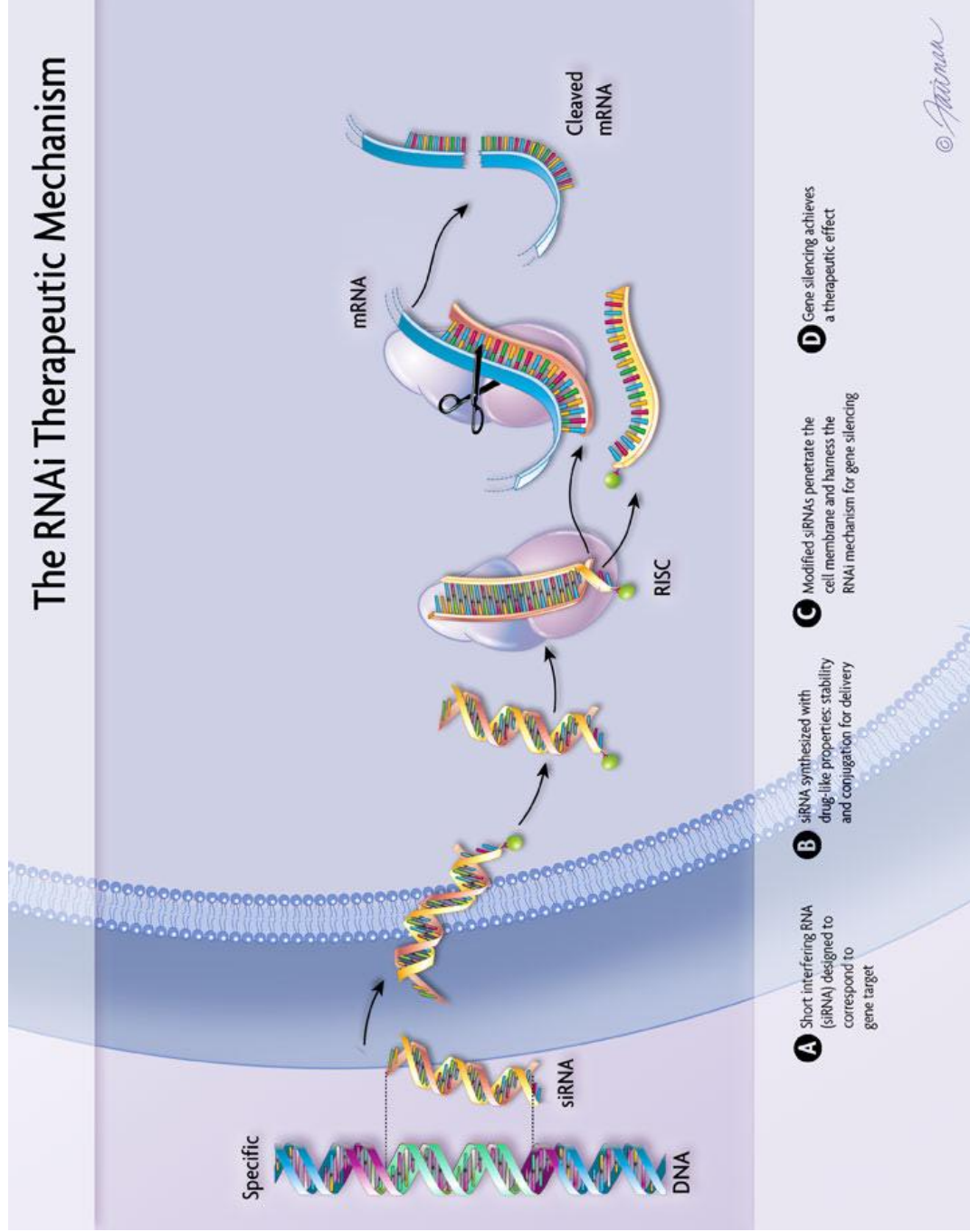
Source: Prepared by casewriters.

Exhibit 5 RNA Interference Mechanism



Source: Company presentation.

Exhibit 6 RNAi Therapeutic Mechanism



Source: Company presentation.

Exhibit 7 Founders of Alnylam Pharmaceuticals

Phillip Sharp, PhD	Institute professor and director, Koch Center for Integrative Cancer Research from 1985 to 1991, MIT. He was director of the McGovern Institute for Brain Research at MIT from 2000 to 2004. Sharpe was a codiscoverer of gene splicing and shared the 1993 Nobel Prize in Physiology or Medicine with Richard Roberts for “the discovery that genes in eukaryotes are not contiguous strings but contain introns, and that splicing messenger RNA to delete those introns can occur in different ways, yielding different proteins from the same DNA sequence.” Cofounder of Biogen and Magen Biosciences.
Dave Bartel, PhD	Professor of Biology, MIT, member of the Whitehead Institute, and investigator of the Howard Hughes Medical Institute. Bartel’s lab was among the early ones to report the existence of hundreds of microRNAs, which regulate the expression of protein-coding genes in animal and plant cells. It then developed the methodology needed to identify which genes each microRNA regulates for both plants and animals.
Paul Schimmel, PhD	Ernest and Jean Hahn Professor of Molecular Biology and Chemistry at the Skaggs Institute for Chemical Biology at the Scripps Research Institute, formerly the John D. and Catherine T. MacArthur Professor of Biochemistry and Biophysics in the Department of Biology at MIT; member, National Academy of Sciences; cofounder of Cubist Pharmaceuticals, Alkermes, and Repligen Corp. Major research activities have concentrated on the decoding of genetic information, with emphasis on the rules of the universal genetic code that are established through aminoacylation reactions catalyzed by a group of enzymes known as aminoacyl tRNA synthetases.
Thomas Tuschl, PhD	After a postdoc at the MIT Whitehead Institute, returned to Max Plank Institute for Biophysical Chemistry. In 2003, Tuschl became a professor at Rockefeller University.
Phillip Zamore, PhD	Gretchen Stone Cook Professor of Biomedical Sciences, UMass Medical School. Extensive pioneering work in RNAi and microRNA.

Source: Compiled by casewriters.

Exhibit 8 Securing the IP Estate

Patent		Background		Alnylam Actions	
Fire and Mello		The Fire and Mello RNA interference patent is based on the pioneering work that the two performed and published in 1998 in Nature for which they obtained the Nobel Prize in 2006. They were the first to identify the trigger mechanism to the RNA interference mechanism to be double-stranded RNA.		Unrestricted, broadly licensed	
		The Tuschl I patent is completely based on research published on <i>Drosophila</i> (fruit fly) embryo extracts where the experiments in mammalian cells were supposedly unsuccessful. The patent did not define the best mode of siRNA (such as the size of the molecule, double-strandedness, <u>unpaired nucleotides</u> at the 3' ends), nor was the invention reduced to practice in mammalian cells. Due to the affiliation of each inventor, Zamore, Tuschl, Sharp, and Bartel, who are all cofounders of Alnylam, the filing of Tuschl I was assigned to four academic institutions: MIT, Whitehead, UMass, and Max Planck, although the ownership interests of UMass are in dispute.		Jointly owned by Max Planck, MIT, Whitehead Institute, and UMass. Licensed to Alnylam, Sirna Therapeutics (Merck), and Cytrx Corp. (further sublicensed to RXi, Inc.).	
Tuschl I					
Tuschl II		Tuschl and colleagues at Max Planck filed a patent on the siRNA invention (known as the Tuschl II invention) with a European priority date of December 1, 2000. In this patent and the subsequent papers published on January 15, 2001, and May 24, 2001, the inventors described the essential elements and requirements of siRNA and experiments that reduced to practice the invention in human cells for the first time. Two essential and unique structural requirements defining the Tuschl II invention are that the small therapeutic siRNA molecules must be double-stranded and contain one or two unpaired nucleotides at the 3' terminal nucleotide ends of the molecule that was otherwise double-stranded. This patent, and subsequent members of this patent family, are exclusively owned by Max Planck and solely licensed to Alnylam Pharmaceuticals.		Exclusively licensed to Alnylam, which has the right to sublicense.	
		Stanley Crooke, the inventor, assigned these patents to Isis Pharmaceuticals, which exploits its expertise in RNA to develop therapeutic drugs. It claims oligomeric compounds, including oligo-ribonucleotides and oligoribonucleosides that contain subsequences of 2-pentoribofuranosyl nucleosides that activate ribonuclease for target mRNA degradation.		In March 2004, Alnylam secured a cross-license with Isis combining Isis's intellectual property and development expertise with Alnylam's intellectual property and research expertise in RNAi therapeutics. This alliance provides Alnylam exclusive access to the Crooke patents that relate to antisense mechanisms and oligonucleotide chemistry for double-stranded RNAi therapeutics, in addition to over 150 chemistry patent issues in the U.S. for modifying double-stranded oligonucleotides.	
Crooke					
Benitec		A little known Australian company, Benitec of St. Lucia in Queensland, was granted a patent on the basis of work done at Queensland's Department of Primary Industries. The work focused on how to deliver RNA strands into a mammalian cell by using a "vector." A DNA construct within the cell gets transcribed into a single strand of RNA, which then folds into a hairpin shape. The cell then recognizes this molecule as a double-stranded RNA. Benitec has the strongest proprietary position in the cellular-expressed double-stranded ribonucleic acid interference (ddRNAi) sector.		In May 2005, Alnylam formed a cross-license with Benitec with terms that offer an option for Alnylam to non-exclusively license Benitec IP for expressed RNAi and synthetic siRNAs in exchange for a reciprocal option for Benitec to non-exclusively license Alnylam-controlled IP in the field of expressed RNAi in addition to options to take up to five Interferex exclusive licenses to pursue synthetic RNAi therapeutics against mutually agreed, specific targets.	

Patent	Background	Alynylam Actions
Nucleonics	Because of the value of the core Benitec patent, a competitor (Nucleonics) has requested the U.S. Patent Office (USPTO) to reexamine the key patent (US 6,573,099 or '099'), in addition to launching two unsuccessful appeals in the U.S courts. Nucleonics is a Pennsylvania-based, privately held company focused on gene therapy applications of RNAi. The Nucleonics IP estate includes over 100 active patent filings, including 16 that have been granted worldwide and 6 that have been granted in the U.S., Europe, or Japan, notably, the "Li & Kirby," "Pachuk I," and "Giordano" patent families that cover broad structural features of RNAi therapeutics.	Alynylam acquired Nucleonics IP estate in December 2008. As a result of Alynylam's purchase of Nucleonics' IP estate and Nucleonics' subsequent failure, this litigation suit was settled and dropped by Alynylam.
Kreutzer-Limmer	The Kreutzer-Limmer IP estate currently comprises mainly four patents in Kreutzer-Limmer I and one patent for Kreutzer-Limmer II granted by the European patent office (EPO). In January 2009, following the challenge of Sirna Therapeutics, a wholly owned subsidiary of Merck & Co., the Technical Board of Appeal of the European Patent Office revoked patent EP 1214945 or '945 in the Kreutzer-Limmer I estate because of its broad claims related to RNAi mediating siRNA molecules. However, as of May 2010, the EPO ruled in favor of Alynylam with regard to the '945 patent that comprises broad claims covering the length and certain structural requirements for siRNAs that are important for advancement of RNAi therapeutics. Several other appeals to both the Kreutzer-Limmer I and II estates have been filed in Europe, but all have been upheld.	In an effort to secure access to the Kreutzer-Limmer IP estate in Europe, Alynylam merged with Ribopharma in July 2003. Looking back at the merger with Ribopharma, Maraganore reflected that the criticality of patents that Ribopharma held combined with uncertainty in this new therapeutic technology area enabled Alynylam to strategically acquire and gain access to Ribopharma's IP estate without much competition.

Source: Compiled by casewriters from multiple published sources.

Exhibit 9 Max Planck Amended Complaint, Pages 1-2

Case 1:09-cv-11116-PBS Document 152 Filed 01/22/2010 Page 1 of 41 UNITED STATES DISTRICT COURT FOR THE DISTRICT OF MASSACHUSETTS MAX-PLANCK-GESELLSCHAFT ZUR FÖRDERUNG DER WISSENSCHAFTEN E.V., a corporation organized under the laws of Germany; MAX-PLANCK-INNOVATION GMBH, a corporation organized under the laws of Germany; and ALNYLAM PHARMACEUTICALS, INC., a Delaware corporation, Plaintiffs, v. WHITEHEAD INSTITUTE FOR BIOMEDICAL RESEARCH, a Delaware corporation; MASSACHUSETTS INSTITUTE OF TECHNOLOGY, a Massachusetts corporation; and BOARD OF TRUSTEES OF THE UNIVERSITY OF MASSACHUSETTS, a Massachusetts corporation, Defendants. Civil Action No. 2009-11116-PBS LEAVE TO FILE GRANTED ON JANUARY 21, 2010 REDACTED <u>PLAINTIFFS' FIRST AMENDED COMPLAINT</u> INTRODUCTION 1. This action seeks redress for the wrongful use of and damage to important property rights in a series of inventions. Plaintiff Max-Planck-Gesellschaft zur Förderung der Wissenschaften e. V. ("Max Planck Society") is the owner of valuable intellectual property that is being irreparably harmed by Defendants' wrongful acts. These wrongful acts stem from Defendants' insistence on exploiting intellectual property developed at and owned exclusively by Max Planck Society. Defendants' misconduct has caused Max Planck Society (and Alnylam Pharmaceuticals, Inc., its exclusive licensee) substantial harm and threatens to prevent Max Planck Society from obtaining the full scope of patent protection to which it is entitled for its invention.	Case 1:09-cv-11116-PBS Document 152 Filed 01/22/2010 Page 2 of 41 2. The dispute arises from the prosecution of two sets of patent applications that are directed to inventions that came out of separate lines of research, performed by separate groups of investigators, in the field of RNA interference, a recently discovered process by which genes can be silenced, thereby opening up an entirely new realm of therapies for serious diseases. Dr. Thomas Tuschl is the first named inventor and only common inventor on both sets of patent applications, which are referred to herein as the "Tuschl I" and "Tuschl II" patent applications. 3. The Tuschl I applications were jointly assigned by the four named inventors—Dr. Thomas Tuschl, Dr. Phillip Sharp, Dr. David Bartel, and Dr. Phillip Zamore—to Max Planck Society, and to defendants Whitehead Institute for Biomedical Research ("Whitehead"), Massachusetts Institute of Technology ("MIT"), and the Board of Trustees of the University of Massachusetts ("UMass") (although, as discussed below, the assignment to UMass was improper and UMass should not be a co-assignee of the Tuschl I patent family). Pursuant to inter-institutional agreements among the parties, Whitehead assumed responsibility for prosecuting the Tuschl I applications on behalf of the other co-assignees of the applications. In contrast, the Tuschl II applications are owned exclusively by Max Planck Society and, accordingly, Max Planck Society has responsibility for prosecuting the Tuschl II applications. Plaintiff Alnylam Pharmaceuticals, Inc. ("Alnylam") has received an exclusive license to the Tuschl II applications from Max Planck Society, and has received an exclusive license to the Tuschl I patents and applications from Max Planck Society, Whitehead, and MIT—but not UMass. 4. In March 2001, at Whitehead's request, Max Planck Society allowed Whitehead to incorporate into the specification of a Tuschl I application a description of significant aspects of the invention encompassed by the Tuschl II applications. Max Planck agreed to Whitehead's request with the understanding that the Tuschl I and Tuschl II inventions would be patented separately so that the commercial value of the Tuschl II invention would be reserved for Max Planck Society alone. That agreement is confirmed by a written representation by Whitehead and MIT in May 2004 that the Tuschl II information will not be used to support claims in the Tuschl I invention and by an Information Disclosure Statement ("IDS") filed with the United 2174889.1 03 - 2 -
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Source: Max-Planck Gesellschaft zur Förderung der Wissenschaften E.V. Max-Planck-Innovation GmbH, and Alnylam Pharmaceuticals, Inc. v. Whitehead Institute for Biomedical Research, Massachusetts Institute of Technology, and Board of Trustees of the University of Massachusetts, Civil Action No. 2009-11116-PBS.

Exhibit 10 Max Planck Amended Complaint, Pages 3–4

Case 1:09-cv-11116-PBS Document 152 Filed 01/22/2010 Page 3 of 41 States Patent and Trademark Office ("USPTO") in 2005 stating that Defendants do not rely on the Tuschl II information to support then-pending claims in the Tuschl I applications. 5. Beginning in or around 2005, notwithstanding the IDS filed that same year and letter sent in May 2004, in complete disregard of their contractual, common-law, and statutory obligations, Defendants began conspiring with each other to implement a scheme to use the Tuschl II invention to support claims in the Tuschl I applications and thereby take from Max Planck Society the commercial value of the Tuschl II invention. Specifically, Defendants are irreparably compromising Max Planck Society's ownership and Anylam's licensing rights by refusing to concede that the invention rightfully belongs exclusively to Max Planck Society and cannot be patented in the Tuschl I applications. As a direct result of this wrongful use, Plaintiffs have been and will continue to be damaged by Defendants' actions in depriving Plaintiffs of the value of the Tuschl II patent estate. THE PARTIES 6. Plaintiff Max Planck Society is a corporation organized and existing under the laws of Germany with a principal place of business in Munich, Germany. The English translation of the corporation's full name is "Max Planck Society for the Advancement of Science." 7. Plaintiff Max-Planck-Innovation GmbH ("Max-Planck-Innovation"), formerly known as Garching Innovation GmbH, is a corporation organized and existing under the laws of Germany with a principal place of business in Munich, Germany. Max-Planck-Innovation is the technology-transfer arm of Max Planck Society. Max Planck Society and Max-Planck-Innovation are collectively referred to herein as "Max Planck." 8. Plaintiff Anylam is a Delaware corporation with a principal place of business in Cambridge, Massachusetts. 9. Plaintiffs are informed and believe, and on that basis allege, that Defendant Whitehead is a Delaware corporation with a principal place of business in Cambridge, Massachusetts. 2:17-08091-03 - 3 -	Case 1:09-cv-11116-PBS Document 152 Filed 01/22/2010 Page 4 of 41 10. Plaintiffs are informed and believe, and on that basis allege, that Defendant MIT is a Massachusetts corporation with a principal place of business in Cambridge, Massachusetts. 11. Plaintiffs are informed and believe, and on that basis allege, that Defendant UMass is a Massachusetts corporation with a principal place of business in Worcester, Massachusetts. JURISDICTION AND VENUE 12. This Court has jurisdiction over the subject matter of this action pursuant to 28 U.S.C. §§ 1331, 1338(a) and 1367. 13. Venue is proper in this district pursuant to 28 U.S.C. § 1391(b). FACTUAL ALLEGATIONS <i>Max Planck Society and its Technology-Transfer Arm, Max-Planck-Innovation</i> 14. Plaintiff Max Planck Society is a non-profit academic research institution in Germany whose mission is to perform scientific research in the interest of the public good. For more than half a century, it has been an icon of cutting-edge research in the life sciences, the natural sciences and the humanities. In particular, the Max Planck Society takes up new and innovative research areas that many universities are not in a position to accommodate. These interdisciplinary research areas often do not fit into the university organization, or they require more funds for personnel and equipment than those available at universities. The variety of topics in the natural sciences and the humanities at Max Planck Society complements the work done at universities and other research facilities in important research fields of innovation. 15. Plaintiff Max-Planck-Innovation is Max Planck Society's technology-transfer arm and manages Max Planck Society's patenting and commercialization endeavors. Max-Planck-Innovation was created in order to promote the transfer of Max Planck Society's proprietary technology to the marketplace and thereby further Max Planck Society's mission to make its technological advances available to the public. 2:17-08091-03 - 4 -
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Source: Max-Planck Gesellschaft zur Förderung der Wissenschaften E.V. Max-Planck-Innovation GmbH, and Anylam Pharmaceuticals, Inc. v. Whitehead Institute for Biomedical Research, Massachusetts Institute of Technology, and Board of Trustees of the University of Massachusetts, Civil Action No. 2009-11116-PBS, <http://patentdocs.typepad.com/files/max-planck-amended-complaint-public.pdf>.

Endnotes

ⁱ P. C. Zamecnik and M. L. Stephenson, "Inhibition of Rous sarcoma virus replication and cell transformation by a specific oligodeoxynucleotide," *Proceedings of the National Academy of Science USA* 75 (1978): 280–284.

ⁱⁱ C. Napoli, C. Lemieux, and R. Jorgensen, *Plant Cell* 2 (1990): 279–289.

ⁱⁱⁱ Andrew Fire et al., *Nature* 391 (1998): 806–811.

^{iv} Andrew Fire, Steven Kostas, Mary Montgomery, Lisa Timmons, SiQuan Xue, Hiroake Tibara, Samuel E. Driver, and Craig C. Mello, "Genetic Inhibition by Double-Stranded RNA," US Patent No. 6,506,559.

^v S. M. Elbashir, J. Harborth, W. Lendeckel, A. Yalcin, K. Weber, and T. Tuschl, "Duplexes of 21-nucleotide RNAs mediate RNA interference in cultured mammalian cells," *Nature* 411 (2001): 494–498.

^{vi} <http://blog.nerac.com/rnailitigation/2010/07/09/an-updated-summary-of-events-in-this-case>.